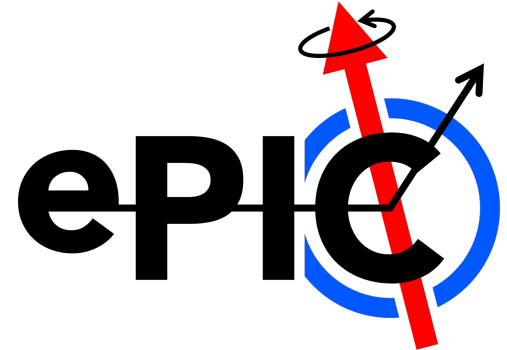


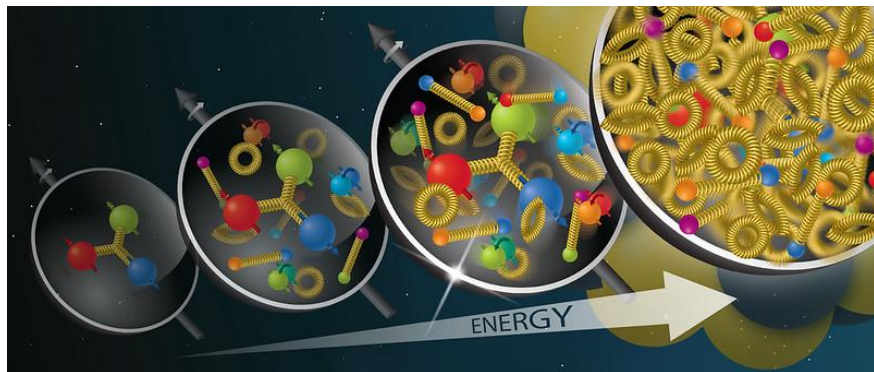
Machine learning based background study for D0 analysis at the EIC

Connie Yang, Deepa Thomas
January 30, 2026

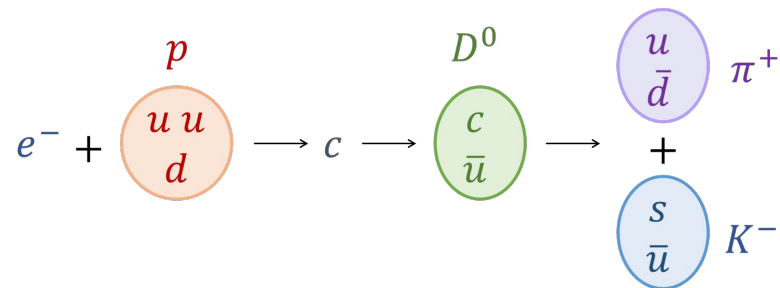


Motivation for heavy flavor analysis at the EIC

- Electron-Ion Collider (EIC) will image the proton and reveal gluon dynamics at high energy
- D0 mesons serve as heavy flavor probes to access the gluons

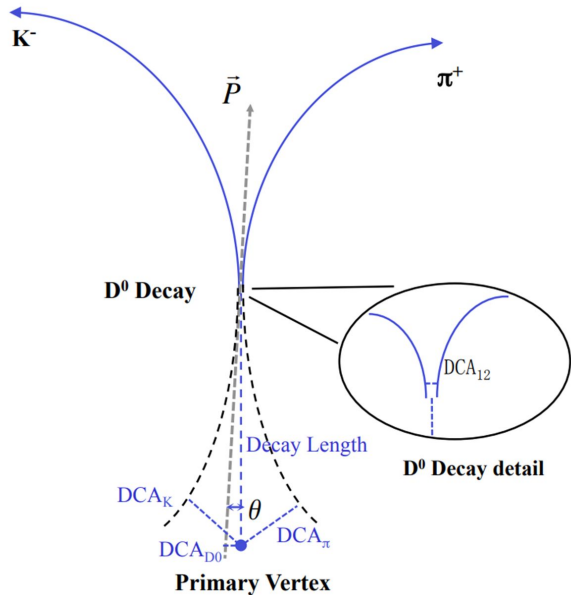


<https://www.bnl.gov/eic/goals.php>

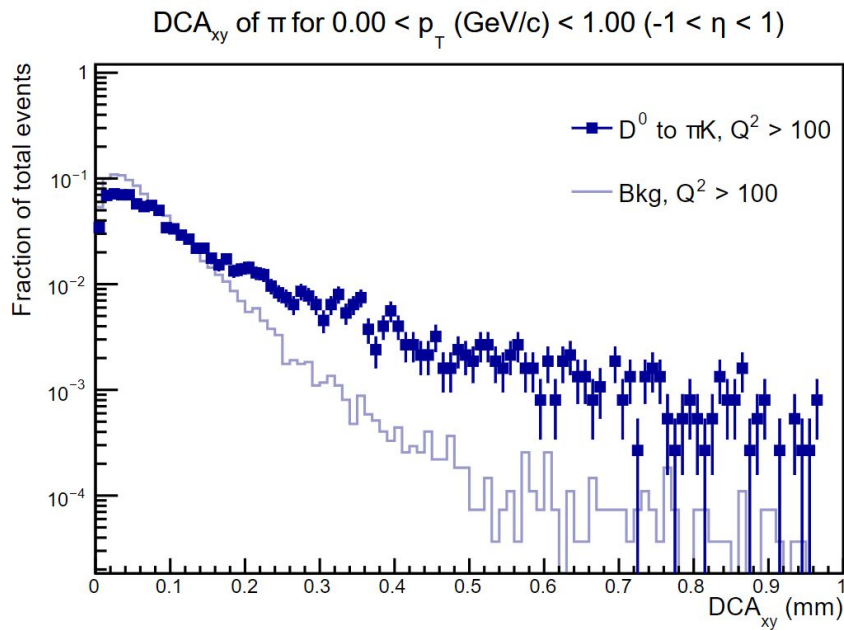


Thesis project

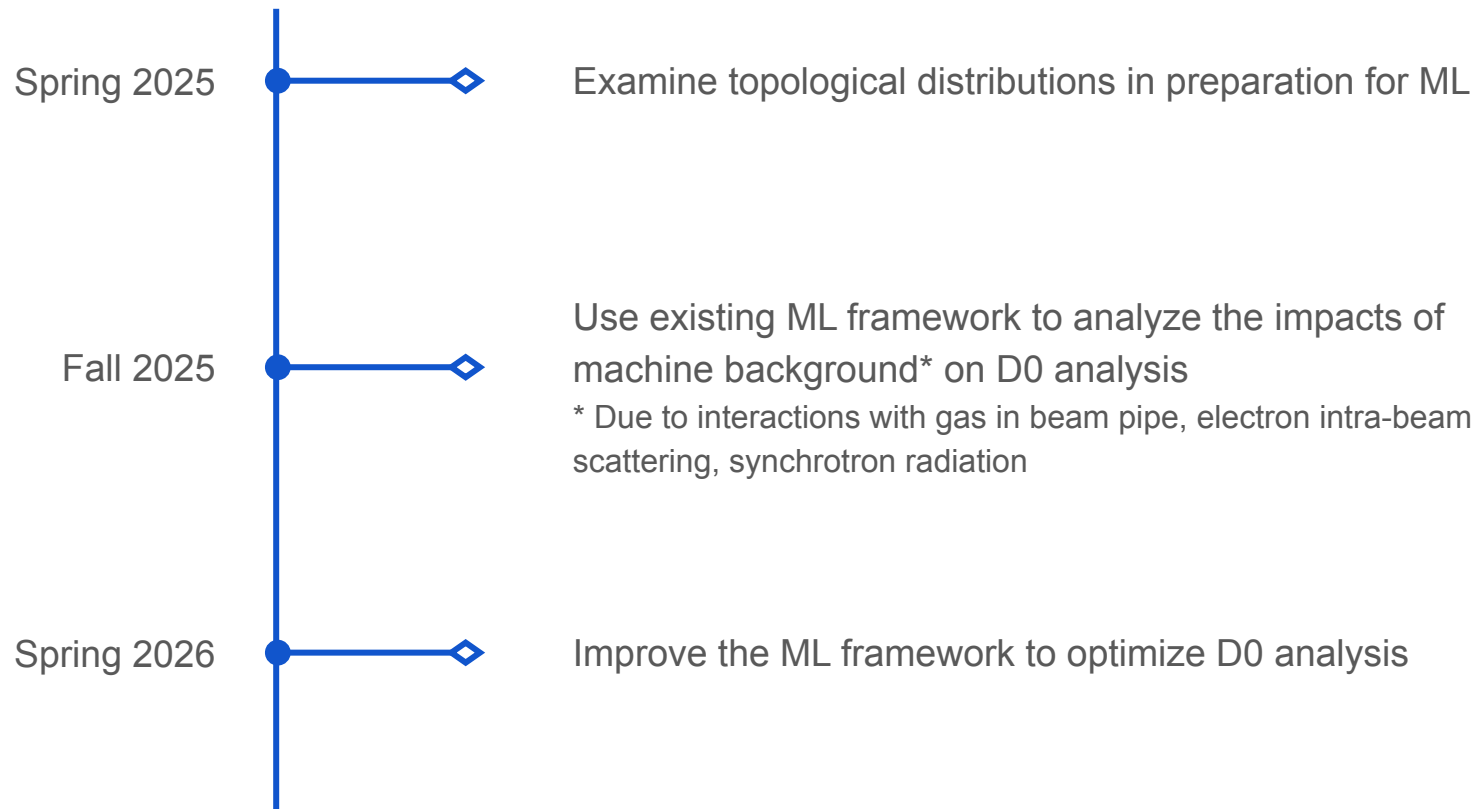
Use machine learning (ML) to separate D0 signals from background based on decay topology.



Phys. Rev. C 99, 034908 (2019)



Project outline



Samples and methods

Simulated data

Without machine background

- 10x100 ep, $Q^2 > 1$, D0 samples $\rightarrow$ signal candidates
- 10x100 ep, $Q^2 > 1$, DIS samples $\rightarrow$ background candidates

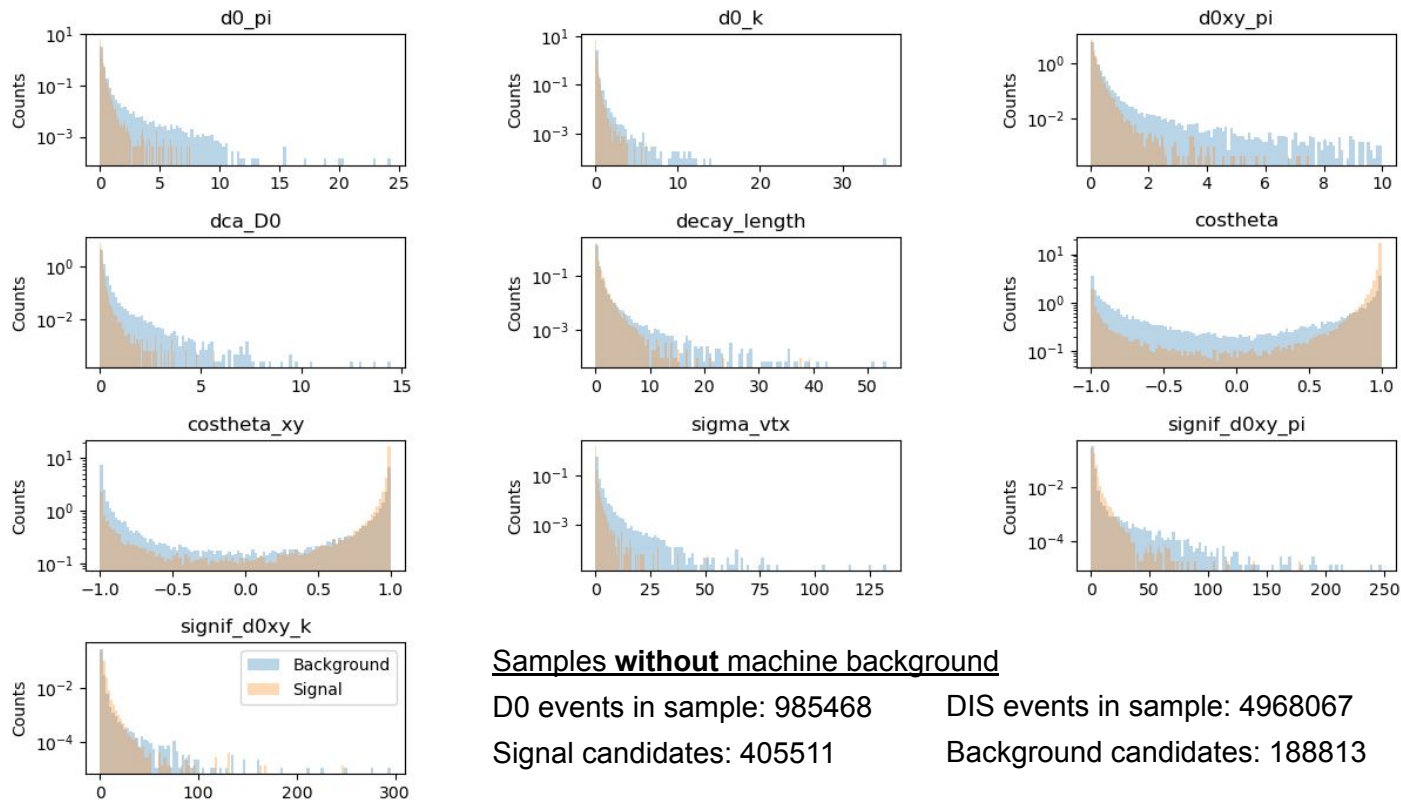
With machine background

- 10x100 ep, $Q^2 > 1$, D0 + machine bkg samples $\rightarrow$ signal candidates
- 10x100 ep, $Q^2 > 1$, DIS + machine bkg samples $\rightarrow$ background candidates

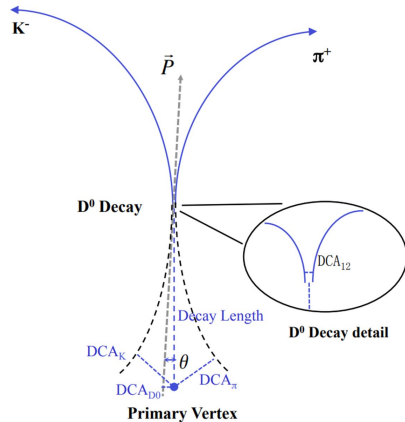
Steps

1. Run analysis to get sig and bkg candidates from samples
(Shyam Kumar, https://github.com/eic/snippets/tree/main/JetsAndHF/SecondaryVertex_Chi2)
2. Train ML models using sig and bkg candidates
(Shyam Kumar, https://github.com/eic/snippets/tree/main/JetsAndHF/ML_HF_Reconstruction)
3. Separate sig and bkg using the trained models
4. Compare D0 invariant mass distributions for samples with and without machine bkg

Without machine bkg, sample statistics and features



Sig and bkg can be distinguished based on topological variables.



Phys. Rev. C 99, 034908 (2019)

Samples without machine background

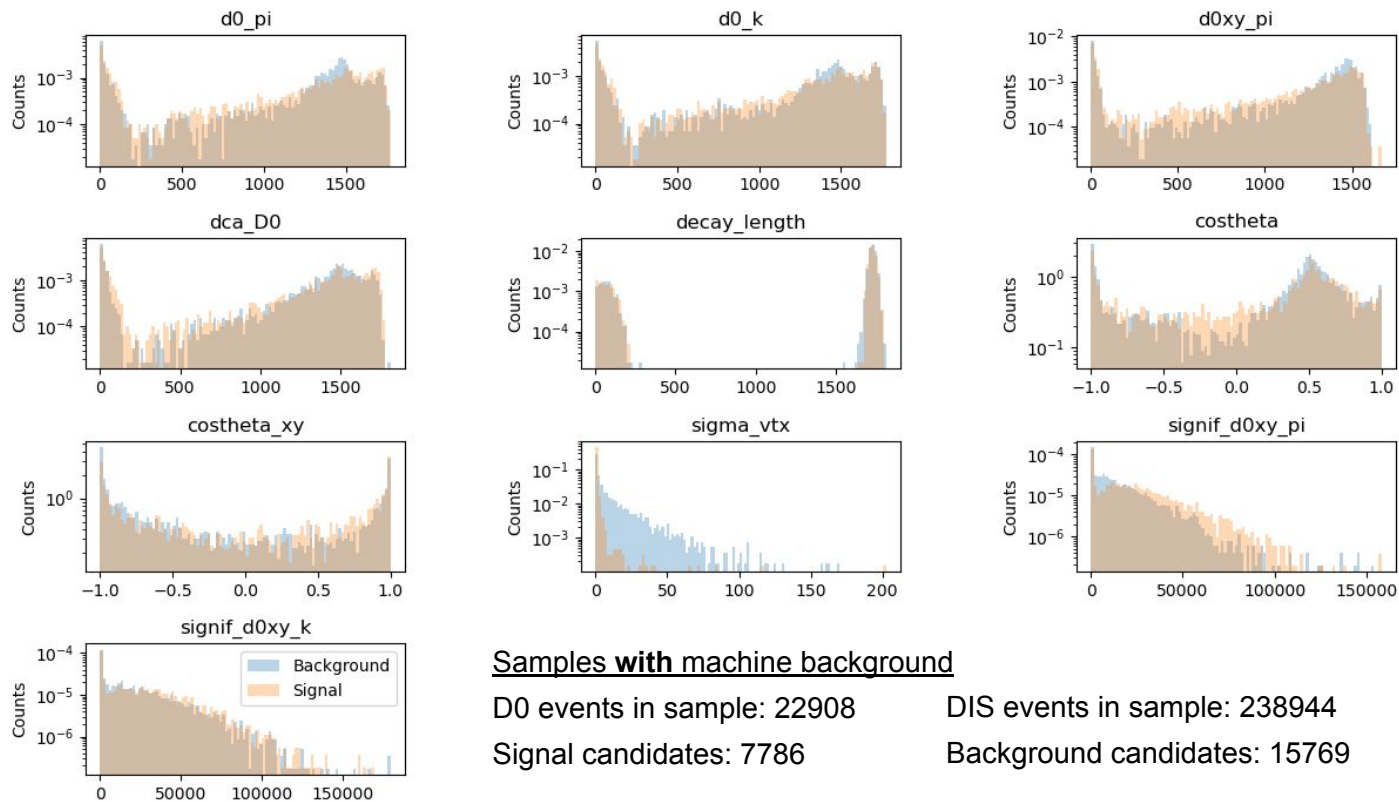
D0 events in sample: 985468

DIS events in sample: 4968067

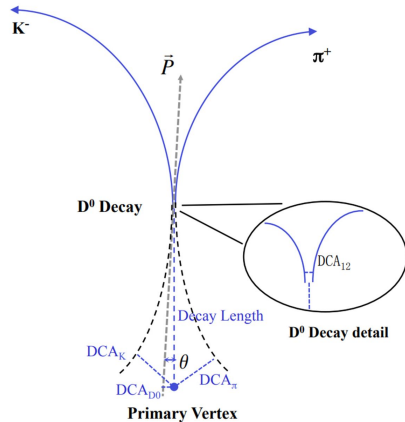
Signal candidates: 405511

Background candidates: 188813

With machine bkg, sample statistics and features



Very large spread in topological distributions with machine bkg.



Samples with machine background

D0 events in sample: 22908

Signal candidates: 7786

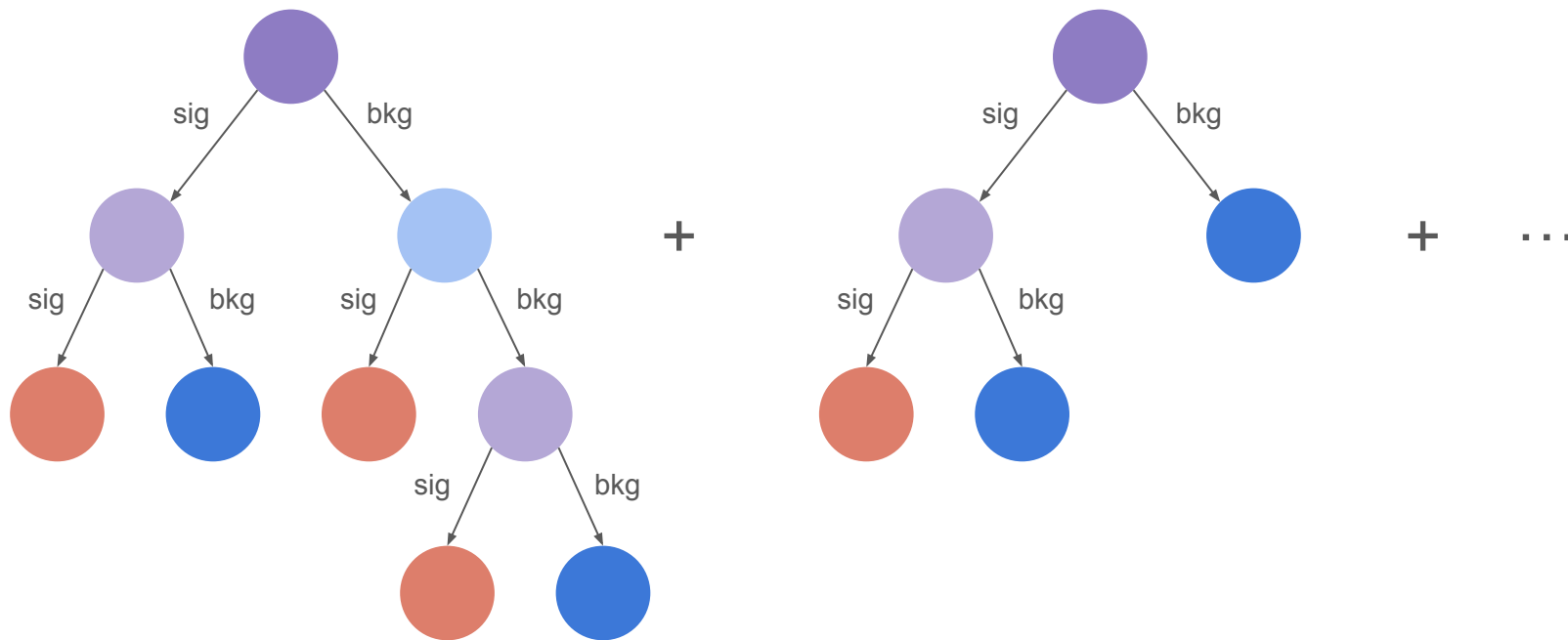
DIS events in sample: 238944

Background candidates: 15769

Phys. Rev. C 99, 034908 (2019)

Boosted decision tree (BDT)

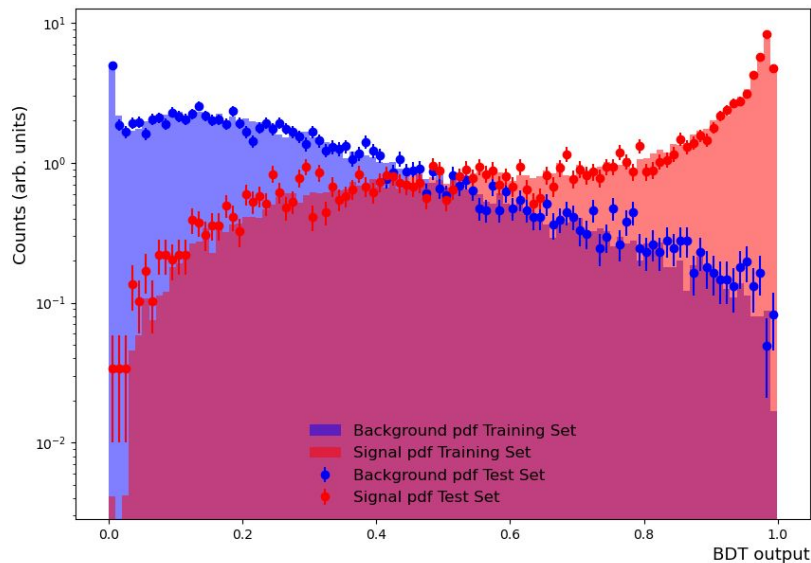
- Decision trees classify candidates as sig or bkg by applying cuts on topological features
- Boosting combines the results of multiple decision trees



Without machine bkg, BDT results

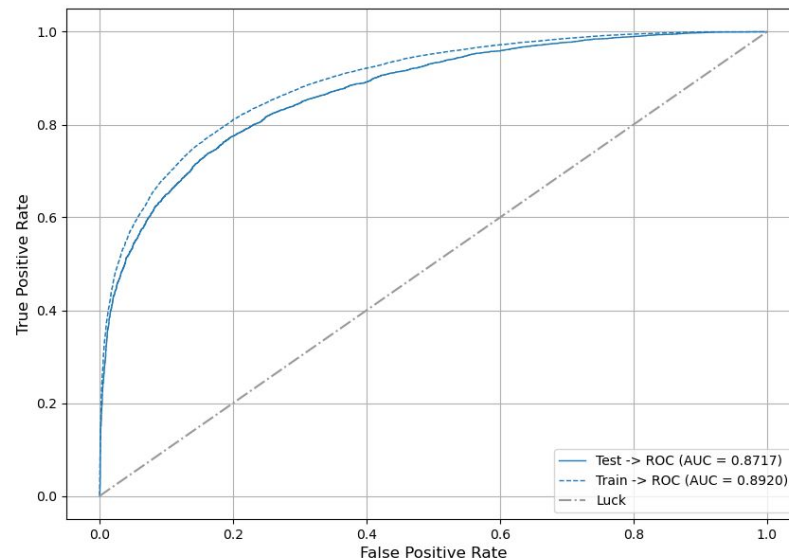
BDT model is well trained and stable.

BDT output describes probability that a candidate is classified as bkg or sig.



Similar results for training and test data.

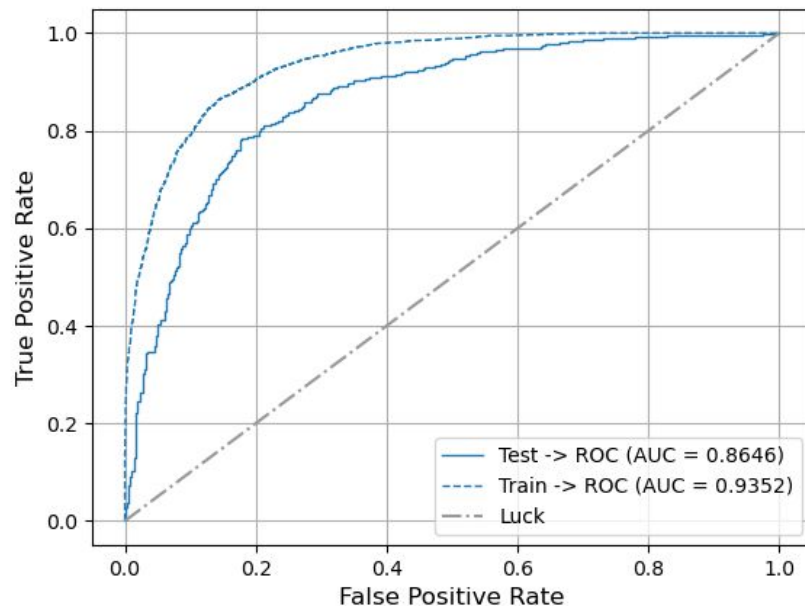
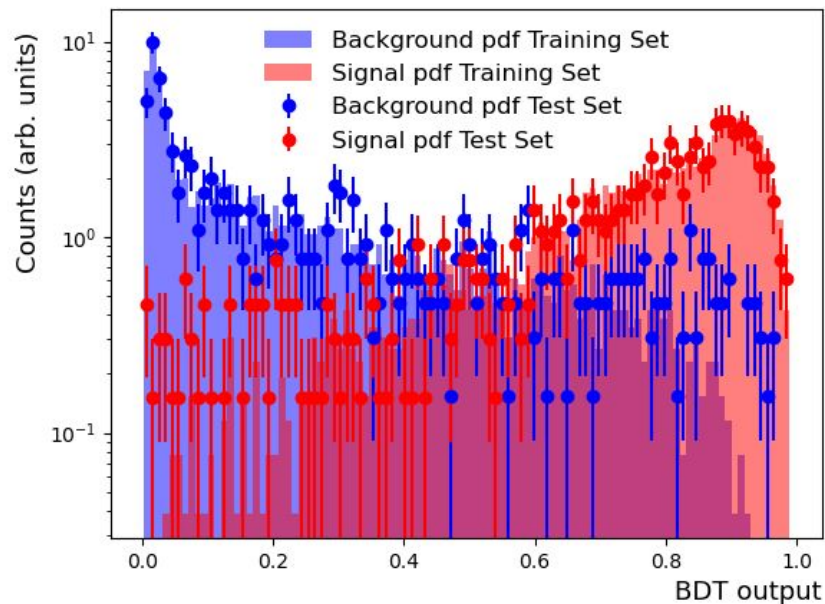
ROC AUC describes the ability to correctly classify signals and background.



Good ROC AUC ~ 0.9 .

With machine bkg, BDT results

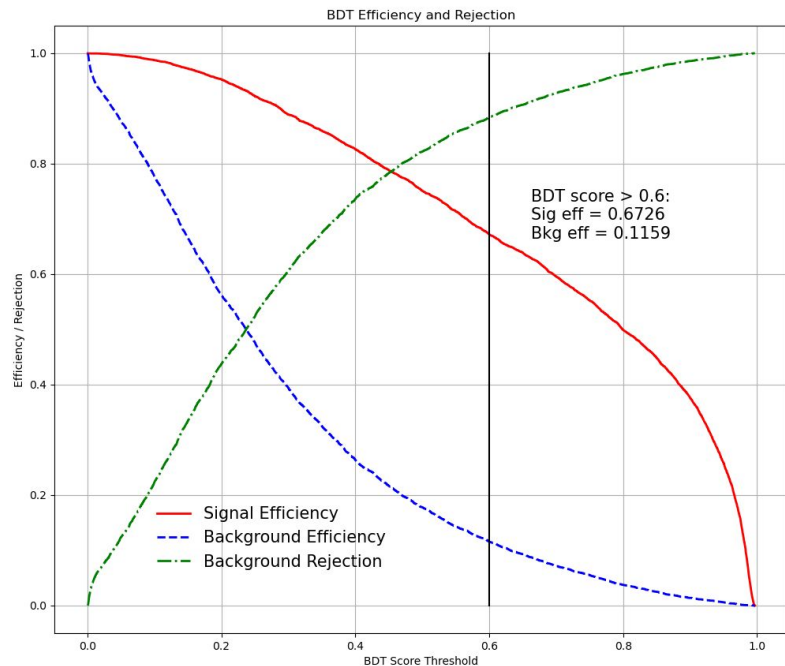
Worse performance for data with machine bkg, BDT training requires further optimization.



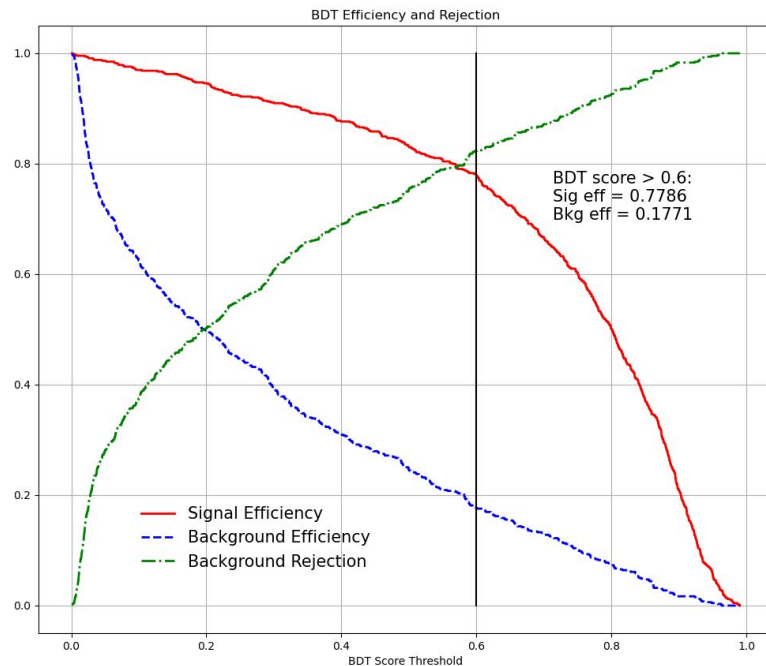
BDT efficiencies

Higher sig eff and bkg eff with machine bkg.

Without machine bkg

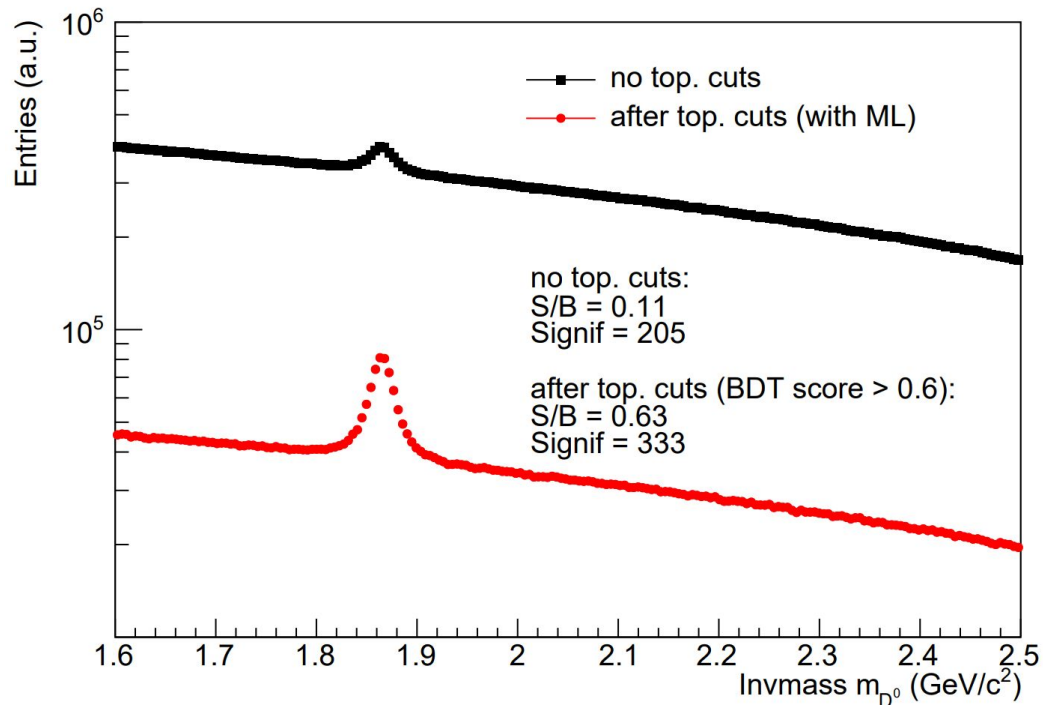


With machine bkg



Without machine bkg, before and after top. cuts

Without machine bkg ($-2 < y < 3$, $pt > 1$)



Topological cuts improve S/B and Signif.

Nevents D0:

$$985468 * 1821.83 \sim 1.80 \times 10^9$$

Nevents DIS: 4968067

$$N_{\text{resampled}} = \frac{N_{\text{candidates}}}{N_{\text{events}}} \times 4.72162e9$$

Expected Nevents
for 10 fb⁻¹

$$N_{\text{resampled}}^{\text{ML}} = N_{\text{resampled}} \times \text{eff}(\text{BDT} = 0.6)$$

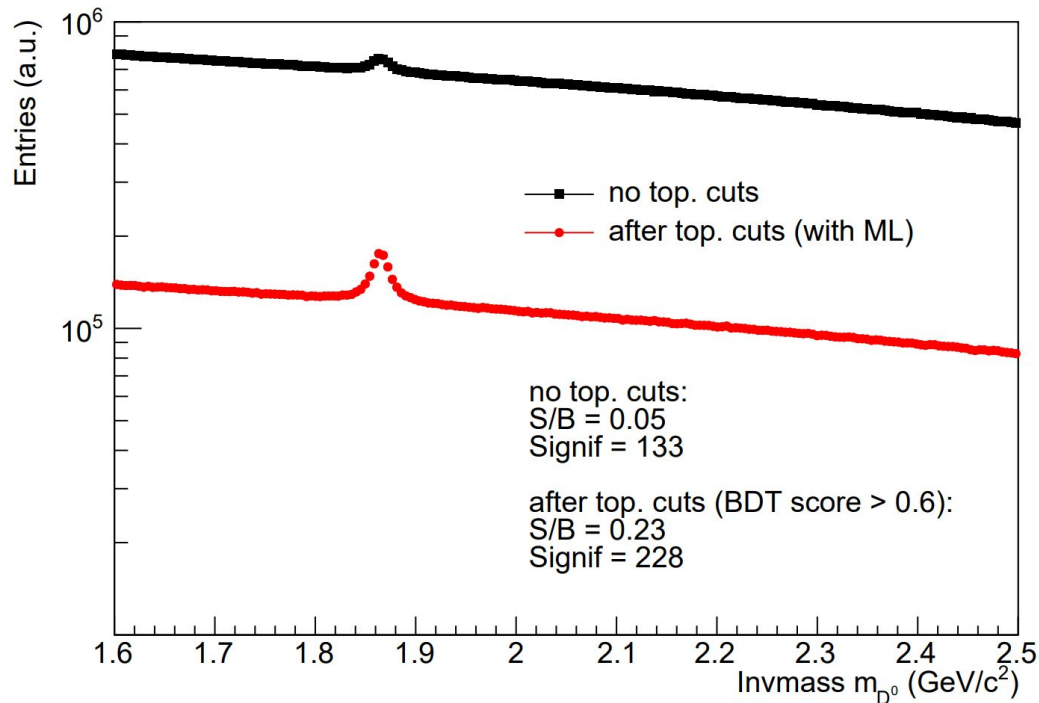
Signal peak: $1.84 < \text{Invmass} < 1.89$

$$S/B = N_{\text{resampled}}^{\text{sig}} / N_{\text{resampled}}^{\text{bkg}}$$

$$\text{Signif} = N_{\text{resampled}}^{\text{sig}} / \sqrt{N_{\text{resampled}}^{\text{sig}} + N_{\text{resampled}}^{\text{bkg}}}$$

With machine bkg, before and after top. cuts

With machine bkg ($-2 < y < 3$, $pt > 1$)



Higher bkg baseline, reduction in S/B and Signif with machine bkg.

Nevents D0:

$$22908.0 * 1821.83 \sim 4.17 \times 10^7$$

Nevents DIS: 238944

$$N_{\text{resampled}} = \frac{N_{\text{candidates}}}{N_{\text{events}}} \times 4.72162e9$$

Expected Nevents
for 10 fb^{-1}

$$N_{\text{resampled}}^{\text{ML}} = N_{\text{resampled}} \times \text{eff}(\text{BDT} = 0.6)$$

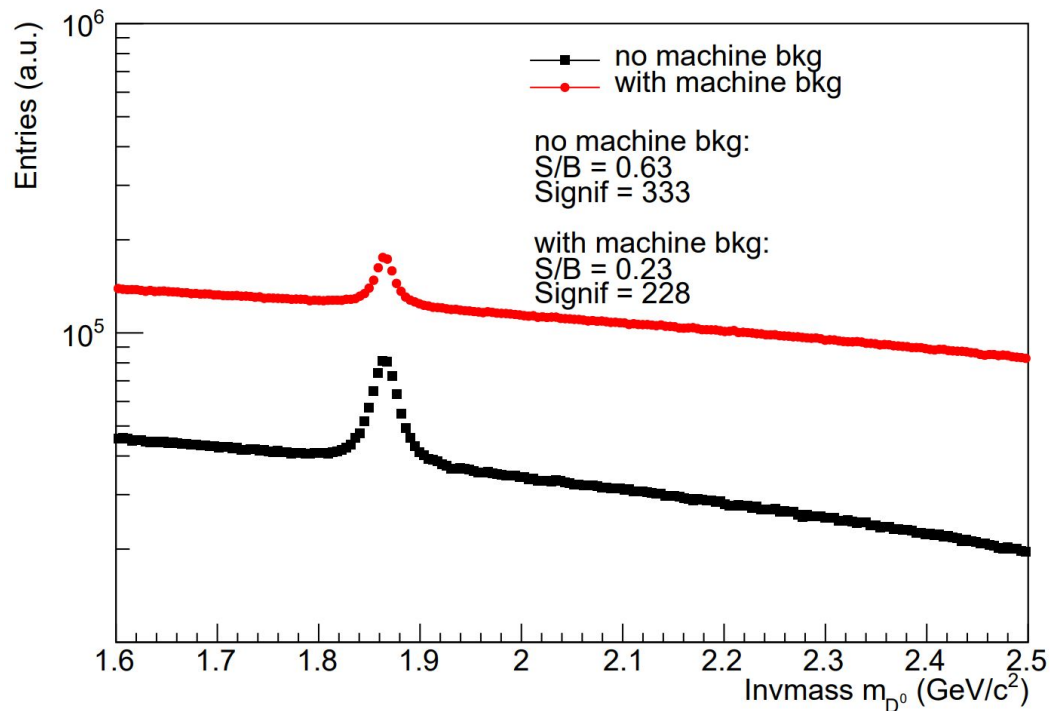
Signal peak: $1.84 < \text{Invmass} < 1.89$

$$S/B = N_{\text{resampled}}^{\text{sig}} / N_{\text{resampled}}^{\text{bkg}}$$

$$\text{Signif} = N_{\text{resampled}}^{\text{sig}} / \sqrt{N_{\text{resampled}}^{\text{sig}} + N_{\text{resampled}}^{\text{bkg}}}$$

Summary and outlook

After top. cuts (BDT score > 0.6 , $-2 < y < 3$, $pt > 1$)



- Trained BDT model on D0 and DIS samples
- Applied topological cuts
- Obtained D0 invariant mass distributions with and without machine bkg
- Observed reduction in S/B and Signif with machine bkg

Next steps

- Optimize BDT training
- Examine primary vertex distribution for samples with machine bkg

References and acknowledgements

October ep simulations

With machine background

D0: /volatile/eic/EPIC/RECO/25.10.4/epic_craterlake/Bkg_Exactly1SignalPer2usFrame/SIDIS/D0_ABCONV/pythia8.306-1.1/10x100/q2_1/hiDiv/

DIS: /volatile/eic/EPIC/RECO/25.10.4/epic_craterlake/Bkg_1SignalPer2usFrame/DIS/NC/10x100/minQ2=1

Without machine background

D0: /volatile/eic/EPIC/RECO/25.10.3/epic_craterlake/SIDIS/D0_ABCONV/pythia8.306-1.1/10x100/q2_1/hiDiv/

DIS: /volatile/eic/EPIC/RECO/25.10.0/epic_craterlake/DIS/NC/10x100/minQ2=1

Analysis framework for sig and bkg candidates:

https://github.com/eic/snippets/tree/main/JetsAndHF/SecondaryVertex_Chi2

Machine learning framework: https://github.com/eic/snippets/tree/main/JetsAndHF/ML_HF_Reconstruction

Based on hipe4ml: <https://doi.org/10.5281/zenodo.5070131>

Thanks to Deepa Thomas, Shyam Kumar, Rongrong Ma

Thank you